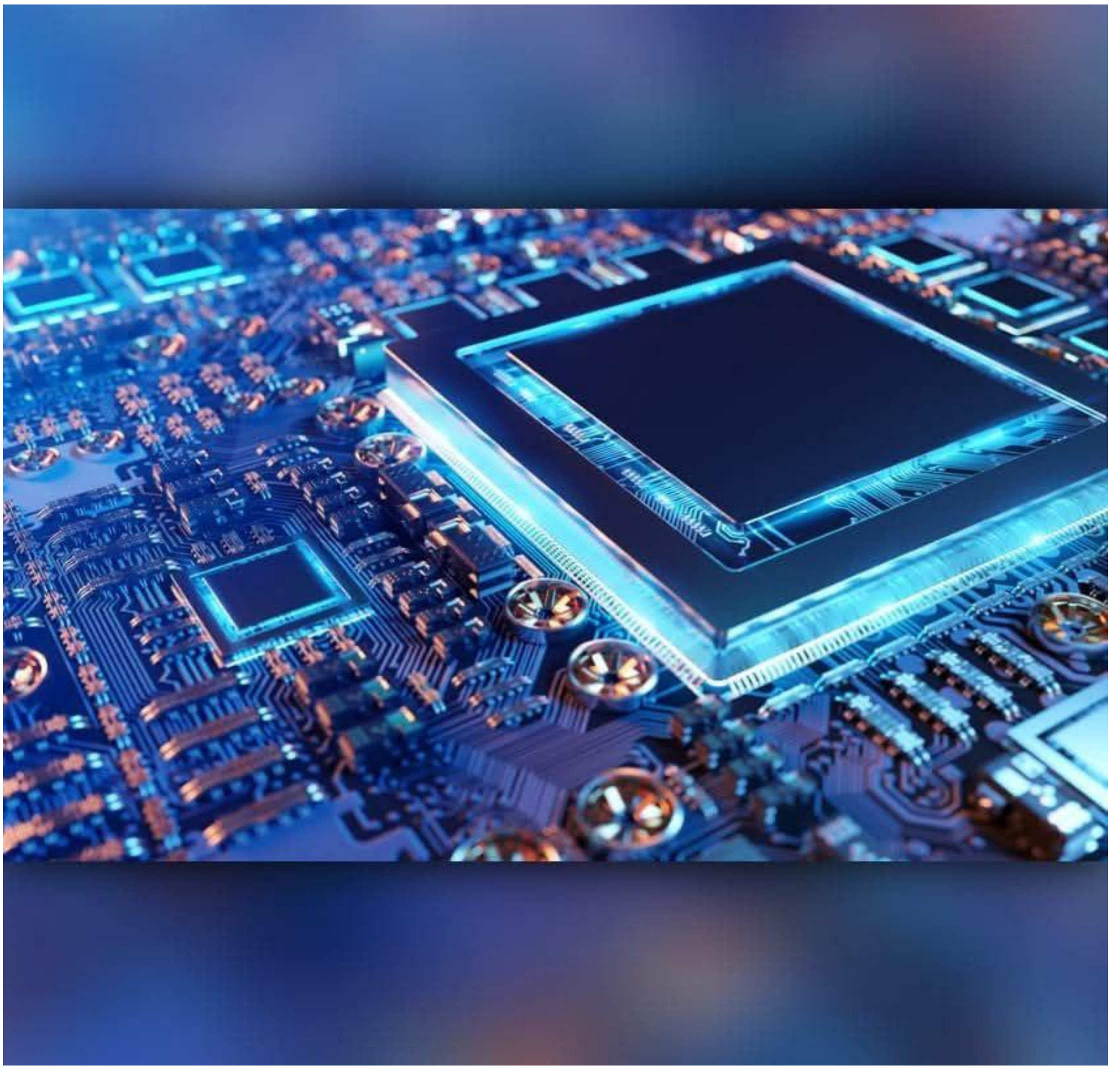
Assignment_One_Extra

Prepared by :

Omar Ahmed Abdelaty



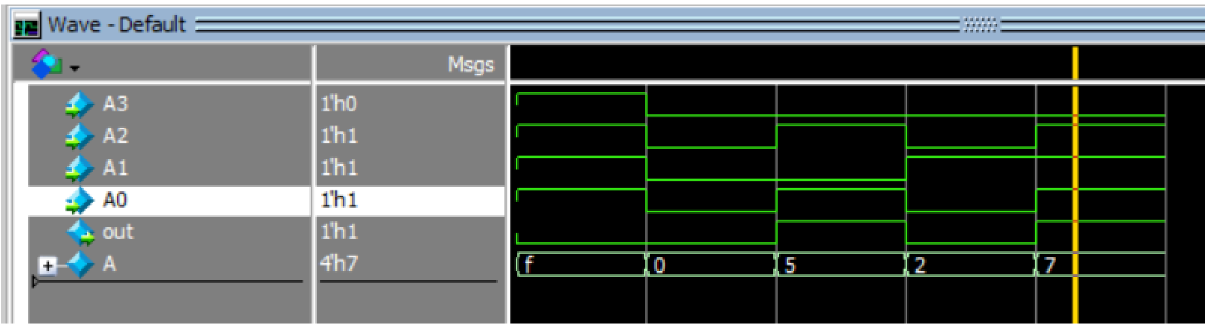
Submitted to: Eng. Kareem Waseem

Question_1

Code

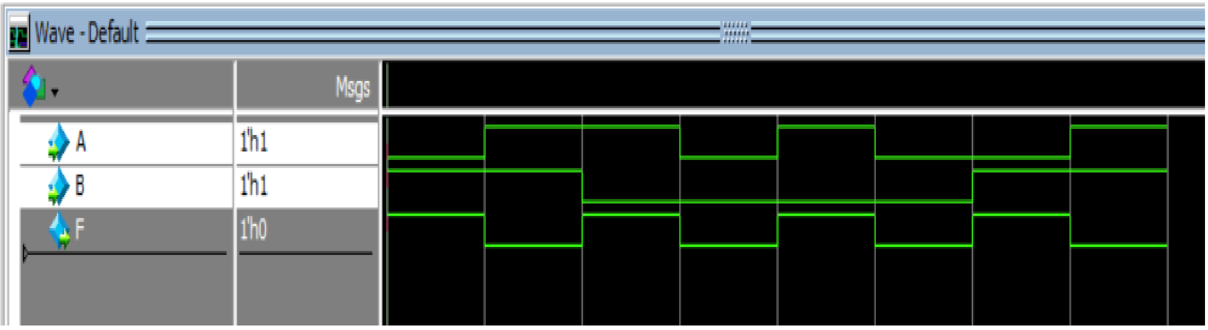
```verilog
module q_1(
    A3, A2, A1, A0, out
);
    input A3, A2, A1, A0;
    output out;

    // Method 1: Using concatenation to form 4-bit vector
    wire [3:0] A;
    assign A = {A3, A2, A1, A0};  // Concatenate individual bits
    assign out = (A > 4'b0010 && A < 4'b1000) ? 1'b1 : 1'b0;

endmodule
```

Simulation

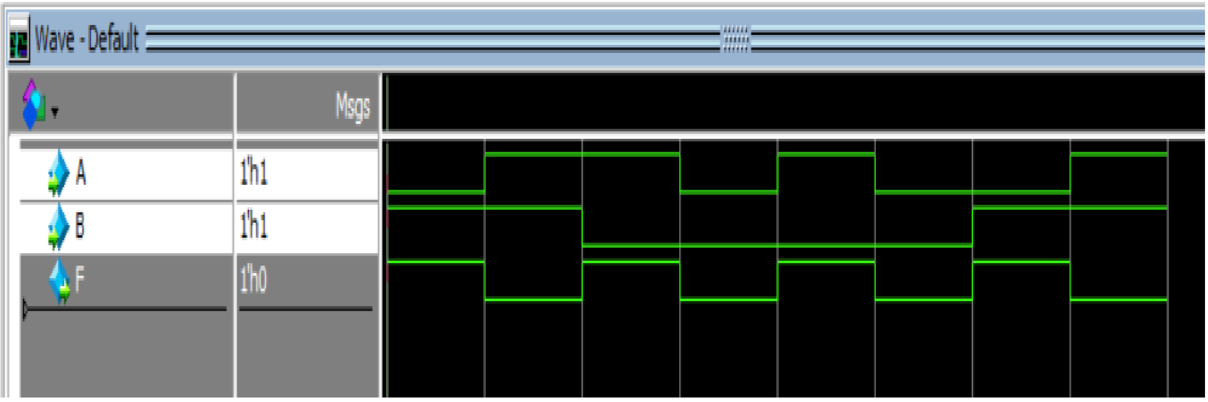


Question_2

Code



Simulation

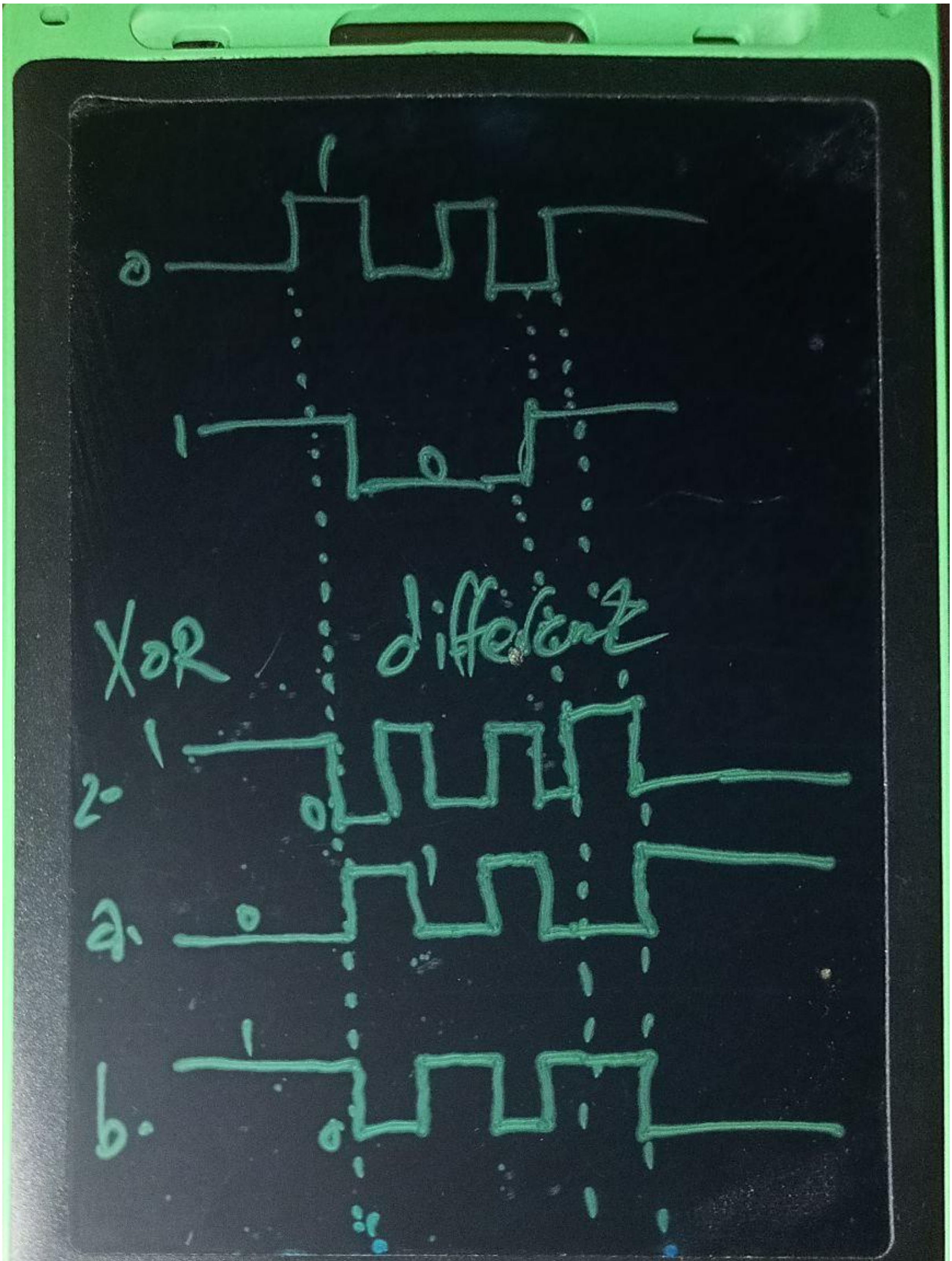


If B held LOW

Then XOR gate will be Buffer

If B held HIGH

Then XOR gate will be inverter

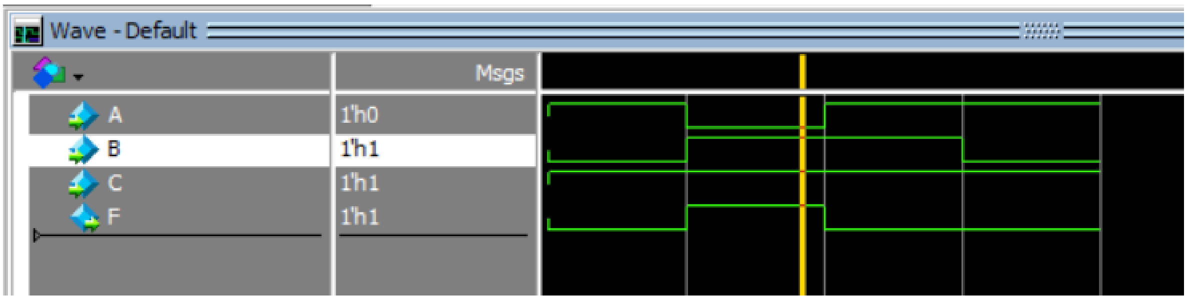

XOR
different
a.
b.

Question_3

Code

```
module q_3(
    A,B,C,F
);

input A,B,C;
output F;

assign F = (A ^ B) & ~(B ^ C) & C ;

endmodule
```

Simulation



The condition for F = 1, we need:(A ^ B) = 1 (A and B different)   and ~(B ^ C) = 1 (B and C same)   and C = 1   so B=C=1 and A=0 only condition

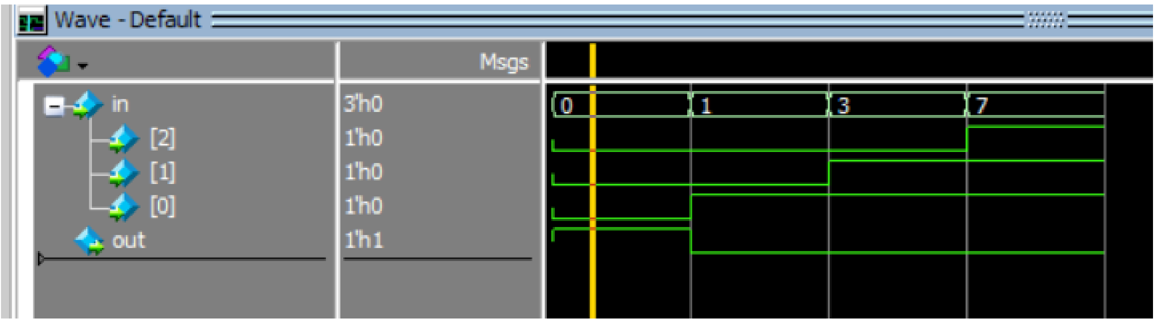
Question_4

Answer is NOR Gate as ~(0 | 0 | 0) = 1

Code

```
module q_4(
    input [2:0] in,
    output out
);

assign out = ~(in[2] | in[1] | in[0]);  // I have used NOR Gate

endmodule
```

Sinmulation

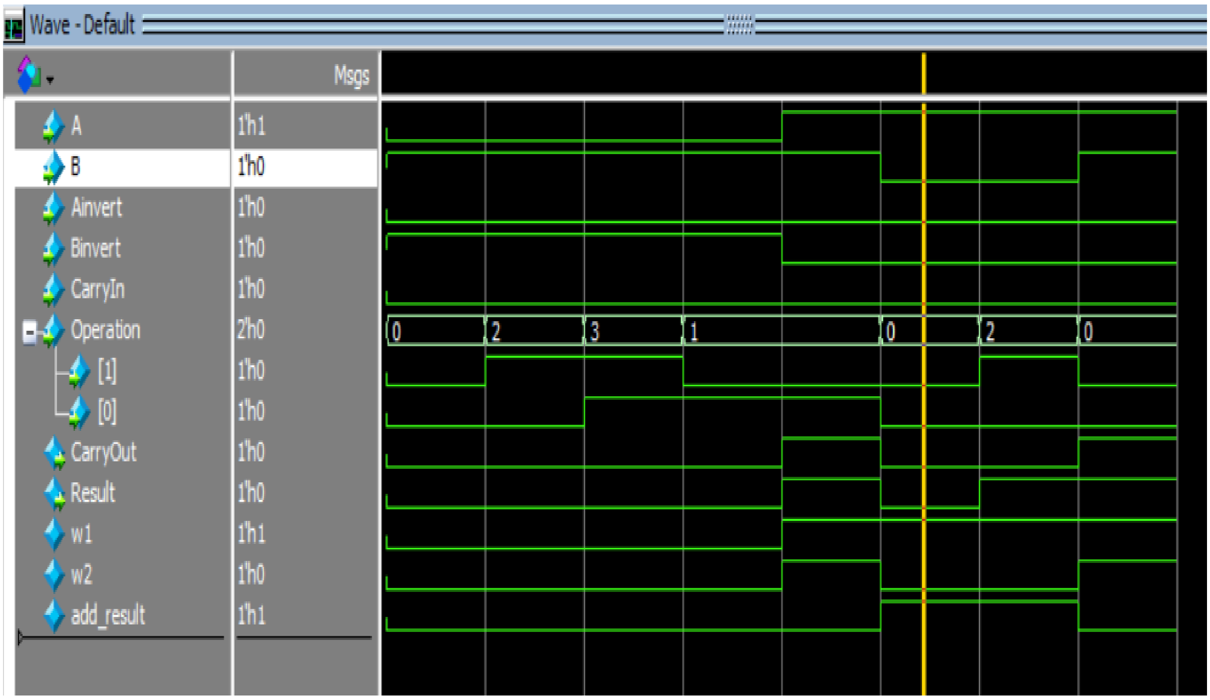


Question_5

Code

```
module one_bit_ALU(
    A,B,Ainvert,Binvert,CarryIn,Operation,CarryOut,Result
);
input A,B,Ainvert,Binvert,CarryIn;
input [1:0] Operation;
output CarryOut,Result;
wire w1,w2;
wire add_result;

assign w1 = (Ainvert==1'b0)? A : (Ainvert==1'b1)? ~A : 1'b0;
assign w2 = (Binvert==1'b0)? B : (Binvert==1'b1)? ~B : 1'b0;
assign {CarryOut,add_result} = w1 + w2 + CarryIn;

assign Result = (Operation==2'b00)? w1 & w2 :
                (Operation==2'b01)? w1 | w2 :
                (Operation==2'b10)? add_result : 1'b0;

endmodule
```

Simulation



Wave - Default
Msgs
A
1'h1
B
1'h0
Ainvert
1'h0
Binvert
1'h0
CarryIn
1'h0
Operation
2'h0
[1]
1'h0
[0]
1'h0
CarryOut
1'h0
Result
1'h0
w1
1'h1
w2
1'h0
add_result
1'h1
0
2
3
1
0
2
0